

Homework 2: Metallic materials

Year 9 Science · about 40 minutes

Section A: A Section B: M / E

Tutor's working, not an official mark scheme

Section A

1 Write the meaning of each word. [A]

(a) density

how much mass is in a given volume (mass divided by volume)

(b) alloy

a mixture of two or more elements, at least one of which is a metal

(c) corrosion

a chemical reaction of a metal with substances around it, such as oxygen and water

2 Name the heat treatment. [A]

(a) A blacksmith wants a horseshoe to be soft, so it is easy to shape.

annealing

(b) The edge of a chisel must be very hard.

quenching

3 Write the word equations. [A]

(a) lead + oxygen $\rightarrow$ lead oxide

(b) sodium + hydrochloric acid $\rightarrow$ sodium chloride + hydrogen

Quick review

4 Name the property of metals that makes each one a good choice. [A]

(a) the copper base of a saucepan

good conductor of heat

(b) gold contacts inside a phone

conducts electricity, unreactive

(c) a steel bridge cable

strong

Section B

5 A gold-coloured crown has a mass of 965 g and a volume of 50.0 cm³. The density of pure gold is 19.3 g/cm³. Is the crown pure gold? Show your working. [M]

density = $\frac{965 \text{ g}}{50.0 \text{ cm}^3} = 19.3 \text{ g/cm}^3$ This matches pure gold, so it could be pure gold.

6 Aircraft bodies are made of an aluminium alloy, not steel. Use the data to explain why.

[E]

Material	Density (g/cm ³)	Strength
aluminium alloy	2.8	high
steel	7.8	very high

The aluminium alloy is strong enough, and its density (2.8) is much lower than steel's (7.8), almost three times less.

A lighter aircraft needs less fuel to fly. Steel is stronger, but it would make the aircraft far too heavy.

7 The steel exhaust pipe of a car rusts faster than the steel body of the car. Explain why.

[M]

The exhaust pipe gets hot, and the burning petrol makes water, so the pipe has water, oxygen and a high temperature. A raised temperature speeds up rusting.

8 A student adds the same mass of four metals to the same volume of dilute acid and measures the temperature rise.

[E]

Metal	magnesium	zinc	iron	copper
Temperature rise (°C)	18	7	3	0

(a) Put the metals in order of reactivity, most reactive first.

magnesium, zinc, iron, copper

(b) Use the data to explain your order, and name two things the student kept the same.

The more reactive the metal, the faster it reacts with the acid, so it gives out more heat: magnesium raises the temperature most (18 °C) and copper does not react at all (0 °C). Kept the same: mass of metal, volume of acid (also concentration of acid, starting temperature).

9 Sam writes about alloys and reactions.

[E]

Sam's answer

1. Steel is harder than pure iron because all the atoms in steel are the same size, so the layers slide easily.
2. magnesium + hydrochloric acid makes magnesium chloride + water
3. Bronze is an alloy, so it is a mixture of metals.
4. When copper is added to dilute hydrochloric acid, bubbles of hydrogen are given off.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. Steel has atoms of different sizes (carbon), which disrupt the layers so they cannot slide easily (L13). 2. Metal + acid gives hydrogen, not water: magnesium chloride + hydrogen (L15). 3. Copper is below the acids in the reactivity series, so there are no bubbles: no reaction (L16).

Also set: Metals booklet p.10–11 (draw the solder graph, then answer Q1–2) and the crossword on p.16.